

Sina Barazandeh

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Pittsburgh
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Education

- 2024 – 2029 **Ph.D. Computational Biology**, School of Computer Science, Carnegie Mellon University, Pittsburgh, PA
Advisor: [Dr. Jose Lugo-Martinez](#)
Relevant Coursework: Machine Learning (10-701), LLM Systems (11-868), Introduction to Computational Structural Biology
- 2021 – 2024 **M.Sc. Computer Engineering**, Bilkent University, Ankara, Turkey
Thesis: Generative Models for Generating and Optimizing Biological Sequences
Advisor: [Dr. A. Ercument Cicek](#) **GPA:** 3.9/4.0
- 2016 – 2021 **B.Sc. Computer Engineering**, Shiraz University, Shiraz, Iran
Thesis: Implementation of Neural Collaborative Filtering
Advisor: Dr. Reza Boostani **GPA:** 3.5/4.0 (16.63/20)

Research Interests

- Foundation Models
- Machine Learning
- Deep Learning
- Computational Biology
- Bioinformatics
- Drug Design
- Reinforcement Learning
- Protein Engineering
- Laboratory Automation

Current Projects

- 2025 Reinforcement Learning for Senescence Biomarker Discovery
Manuscript in Preparation
- 2025 Evolution Informed Structural Subgraph-Based Protein Representations Outperform Foundation Models
Manuscript in Preparation
- 2025 Developing Agentic AI using RL and LLMs for Large-scale Automated Labs
Work in progress

Research Experience

- 2024 – Present **Graduate Research Assistant**
Carnegie Mellon University, School of Computer Science
Advisor: [Dr. Jose Lugo-Martinez](#)
• Biomarker Discovery, Protein Representations, and Automation of Science.
- 2021 – 2024 **Research Assistant**
Bilkent University, Department of Computer Engineering ([CicekLab](#))
Advisor: [Dr. A. Ercument Cicek](#)
• Developed GANs and LLMs for biological sequence generation and optimization
- 2019 – 2020 **Undergraduate Research Assistant**
Shiraz University, Department of Computer Engineering
Advisor: Dr. Reza Boostani
Project: EEG to MEG signal translation using LSTM Neural Networks

Research Publications

- 2026 **Sina Barazandeh**, Gun Kaynar, Arth Banka, Joshua Kangas, Carl Kingsford, Jose Lugo-Martinez. **CloudBench**: An Open-Framework Simulator and Benchmark Generator for Programmable Cloud Laboratories” To be submitted to *Nature Communications*.
- 2026 Ozden, Furkan, **Sina Barazandeh**, Dogus Akboga, Sobhan Shokoueian Tabrizi, Urartu Ozgur Safak Seker, and A. Ercument Cicek. **”RNAGEN**: A generative adversarial network-based model to generate synthetic RNA sequences to target proteins.” In Press at *ChemBioChem*.
- 2025 Shukueian Tabrizi, Sobhan, **Sina Barazandeh**, Helyasadat Hashemi Aghdam, and A. Ercument Cicek. **”RNAtranslator**: Modeling protein-conditional RNA design as sequence-to-sequence natural language translation.” *PLOS Computational Biology* 21, no. 10 (2025): e1013541.
- 2025 **Barazandeh, S.**, Furkan Ozden, Ahmet Hincer, Urartu Ozgur Safak Seker, and A. Ercument Cicek. **”UTRGAN**: Learning to generate 5' UTR sequences for optimized translation efficiency and gene expression.” *Bioinformatics Advances* 5, no. 1 (2025): vbaf134.
- 2018 Asali, Ehsan, Farzin Negahbani, Saeed Tafazzol, Mohammad Sadegh Maghareh, Shahryar Bahmeie, **Sina Barazandeh**, Shokoofeh Mirian, and Mahta Moshkelgosha. ”Namira Soccer 2D Simulation Team Description Paper 2018.” *RoboCup 2018*, Montreal, Canada

Skills

- **Programming Languages**: Python, R, Java, C/C++, Javascript, Elixir, Verilog
- **Machine Learning tools**: PyTorch, Keras, Scikit-Learn, AWS Strands
- **Foundation Models**: ESM-2, AlphaFold, Boltz, ProtT5, RoseTTAFold, DNABERT
- **Bioinformatics Tools**: BLAST, HMMER, Clustal, PyMOL, Galaxy, BioPython, NCBI Tools
- **Databases**: PDB, UniProt, NCBI, BioMart, Bioconductor
- **Computing Infrastructure**: SLURM, HPC Clusters
- **GPU Computing**: CUDA basics, Multi-GPU Training, PyTorch distributed training
- **Software Development**: Git, Docker, Singularity, CI/CD
- **Graphics & Image Processing**: OpenGL, OpenCV

Conferences and Presentations

- 2026 **SciFM: Foundation Models and Agentic AI for Science - Chicago, IL**
- 2025 **SenNet Annual Meeting - Washington, DC**
- 2025 **PathVisions - San Diego, CA**
- 2023 **RECOMB - Istanbul, Turkey** (Volunteer)
Poster Presentation: **UTRGAN**
- 2023 **RECOMB-SEQ - Istanbul, Turkey**
Poster Presentation: Using Minimizer Interarrival Distances for Read-Until Human Read Detection from Blood Samples Sequenced by Oxford Nanopore
- 2023 **HIBIT - Ankara, Turkey**
Oral Presentation: **RNAGEN(Second Best Presentation Award)**
- 2019 **Shiraz University** (Invited Presentation)
Introduction to Soccer 2D Simulation

Awards & Honors

2021	Comprehensive Graduate Scholarship, Bilkent University
2021	Best Start-Up in Southern Iran Award (GhasedakApp)
2021	Ranked 1st in Hardware Engineering, Top 10% of Class of 2020
2018	8th Place (out of 32 teams), RoboCup Soccer 2D Simulation League, Montreal, Canada
2018	5th Place (out of 24 teams), Iran Open Soccer 2D Simulation League, Tehran, Iran
2016	Full National Scholarship for Undergraduate CS Program, Shiraz University, Iran

Teaching Experience

2026	Teaching Assistant: Machine Learning for Scientists Carnegie Mellon University Instructor: Dr. Martin Zhang
2025	Teaching Assistant: Computational Medicine Carnegie Mellon University Instructor: Dr. Wei Wu
2022	Teaching Assistant: Artificial Intelligence Bilkent University Instructor: Dr. Ozgur Oguz
2021	Teaching Assistant: Introduction to Machine Learning Bilkent University Instructor: Dr. Aysegul Dundar

Academic Service

2026	Reviewer - Bioinformatics
2025	Subreviewer - ISMB Conference
2024	Subreviewer - RECOMB Conference
2023	Subreviewer - ISCB Conference
2023	Subreviewer - RECOMB-SEQ Satellite Conference
2022	Subreviewer - RECOMB Conference
2022	Subreviewer - ISBRA Symposium
2022	Subreviewer - ACM-BCB Conference

Course Research Projects

2026	Attentions are all you need: Extending FlashAttention and Paged-Attention to Minitorch Instructor: Lei Li
2025	Investigating ProtT5 and ESM Embeddings Disentanglement on Multi-Chain Proteins Instructor: David Ryan Koes
2025	Complex Cellular Circuit Design for Stable Cellular Systems Instructor: James R. Faeder and Keisuke Ishihara
2024	VAE Disentanglement Analysis on Noisy data using Residual Connections Instructors: Maria Florina Balcan and Geoffrey J Gordon
2023	Benchmarking Long-read <i>de novo</i> Genome Assembly Tools for Metagenomics Instructor: Dr. Inanc Birol
2022	Bit-Packing for Contig Content Summarization Applications Instructor: Dr. Inanc Birol
2021	Text-Driven Face Image Manipulation in GANs Course: Generative Networks
2021	Attention-Based Encoder-Decoder Anomaly Detection Course: Deep Learning

Software Engineering Projects

- 2023 **Deep Learning meets Blockchain for Automated and Secure Access Control**
Collaborators: Asma Jodeiri Akbarfam, **Sina Barazandeh**, Deepti Gupta, Hoda Maleki
Deployed deep learning models on Public Amazon Employee Access Challenge dataset for enterprise access management

Entrepreneurial Experience

- 2019-2022 **Co-Founder and Backend Developer**
[Ghasedak Institution](#), Shiraz, Iran
Developed backend infrastructure for a online store for an Amazon-level access to book stores accross the country.

Leadership & Extracurricular Activities

- 2019-2021 **Vice-President**, Interdisciplinary Robotics Association
Shiraz University, Shiraz, Iran
Led initiatives promoting robotics education and organized inter-departmental meetings.

References

- **Dr. Jose Lugo-Martinez**
Assistant Professor, School of Computer Science, Carnegie Mellon University
Email: jugomar@cs.cmu.edu
- **Dr. Carl Kingsford**
Herbert A. Simon Professor of Computer Science, Carnegie Mellon University
Email: carlk@cs.cmu.edu
- **Dr. A. Ercument Cicek**
Associate Professor, Department of Computer Engineering, Bilkent University
Adjunct Professor, Computational Biology Department, Carnegie Mellon University
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